

The Concept of MLEs

- Suppose that $X_1, \dots, X_n$ are i.i.d. with common pdf $f(x; \theta)$, $\theta \in \Omega$.
- Is there a systematic method we can perform statistical inference such as point estimation, confidence intervals, and hypothesis testing?
- In this section, we develop statistical inference (estimation and testing) based on likelihood methods.

- Likelihood function:

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n f(x_i; \theta), \theta \in \Omega.$$

- Log likelihood:

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n \log f(x_i; \theta).$$

- Attention: They are functions of θ !

A Motivation Example

- Let $X_1, \dots, X_n$ denote a random sample from the distribution with pmf $p(x) = \theta^x(1 - \theta)^{1-x}$, $x = 0, 1$, zero elsewhere, where $0 < \theta < 1$. The joint pmf is

$$\theta^{x_1}(1 - \theta)^{1-x_1} \dots \theta^{x_n}(1 - \theta)^{1-x_n} = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}.$$

The likelihood function is

$$L(\theta) = \theta^{\sum_{i=1}^n x_i} (1 - \theta)^{n - \sum_{i=1}^n x_i}.$$

- We might ask what value of θ would maximize the probability $L(\theta)$ of obtaining this particular observed sample $x_1, \dots, x_n$. This maximizing value of θ would be a good estimate of θ because it would provide the largest probability of this particular sample.

- The log likelihood function

$$l(\theta) = \log L(\theta) = \sum_{i=1}^n x_i \log \theta + (n - \sum_{i=1}^n x_i) \log(1 - \theta).$$

$$\frac{dl(\theta)}{d\theta} = \frac{\sum_{i=1}^n x_i}{\theta} - \frac{(n - \sum_{i=1}^n x_i)}{1 - \theta} = 0.$$

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

is called *maximum likelihood estimator* of θ .

- Assumptions: (Regularity Conditions).
 - R0: The pdfs are distinct; i.e., $\theta \neq \theta' \Rightarrow f(x; \theta) \neq f(x; \theta')$.
 - R1: The pdfs have common support for all θ .
 - R2: The point θ_0 is an interior point in Ω .
- **Theorem 6.1.1.** Let θ_0 be the true parameters. Under assumptions R0 and R1,

$$\lim_{n \rightarrow \infty} \mathbb{P}_{\theta_0}[L(\theta_0, \mathbf{X}) > L(\theta, \mathbf{X})] = 1, \quad \text{for all } \theta \neq \theta_0.$$